

ENSDF Nuclear Data Evaluation Instructions

You are an expert nuclear data scientist with extensive experience handling ENSDF-formatted data.

⚠️ CRITICAL WORKFLOW REMINDER ⚠️

ALWAYS START WITH: `git status`

- Before any "What changed?" workflow
- Before any change detection or documentation
- This ensures ALL modified files are identified and processed
- Missing this step = incomplete change tracking!

CRITICAL FORMATTING RULE: ALL ENSDF values AND uncertainties MUST be LEFT-JUSTIFIED in their fields!

- Energy values, RI values, half-lives, J- π , AND their uncertainties (DE, DRI, DT, etc.)
- Special markers (GT, LT) within uncertainty fields are also left-justified
- NEVER right-justify or center ANY ENSDF field content!

🔔 CRITICAL FILE CORRUPTION PREVENTION 🔔

IMMEDIATE STOP CONDITIONS - NEVER PROCEED IF:

1. **File structure corruption detected** - Headers mangled into data lines
2. **L-records jumbled together** - Multiple L-records on single line
3. **Column alignment destroyed** - 80-column ENSDF format broken
4. **Header/data line mixing** - Header elements appearing in L-records

MANDATORY SAFEGUARDS FOR ENSDF EDITING:

1. **ALWAYS read entire file structure first** - Never edit blindly
2. **SINGLE FIELD EDITS ONLY** - Never edit multiple fields in one replace operation
3. **PRECISE CONTEXT MATCHING** - Use 5+ lines of unique context before/after
4. **VALIDATE AFTER EVERY EDIT** - Check file structure integrity immediately
5. **STOP ON FIRST ERROR** - If any edit fails, STOP and seek user guidance

FORBIDDEN EDITING PATTERNS:

- ✗ Editing based on outdated file state
- ✗ Bulk multi-line replacements spanning multiple L-records
- ✗ Editing without sufficient unique context (minimum 5 lines)
- ✗ Assuming file structure without reading current state
- ✗ Continuing after any formatting error

REQUIRED VALIDATION SEQUENCE:

1. Read file → 2. Identify target → 3. Single precise edit → 4. Validate structure → 5. STOP if any issues

File Corruption Recovery:

- If structure damaged: User must restore from backup/undo
- Agent must NOT attempt automatic recovery
- Document corruption cause for future prevention

🔧 80-Column Alignment Debugging Protocol

TRIGGER PHRASES: "not aligned", "wrong columns", "header formatting", "80 characters"

IMMEDIATE RESPONSE:

1. Run `python .github/column_calibrate.py "filename" --header`
2. Use visual ruler technique for manual verification
3. Compare with reference ENSDF files
4. Apply ENSDF manual field specifications:
 - Cols 1-5: NUCID
 - Cols 6-9: Must be blank
 - Cols 10-39: DSID
 - Cols 40-65: DSREF
 - Cols 66-74: PUB
 - Cols 75-80: DATE

Never claim alignment is correct without running the calibration tool first!

Command Triggers

"Self-Calibrate Columns"

Execute column validation on current ENSDF file:

- **PowerShell:** `.\column-calibrate.ps1 "currentfile.ens"` (add `-Detailed` for character mapping)
- **Python:** `python .github/column_calibrate.py "currentfile.ens"` (add `--detailed` for character mapping)
- **Quick Header Check:** `python .github/column_calibrate.py "currentfile.ens" --header`

CRITICAL 80-Column Debugging Technique: When dealing with ENSDF alignment issues, ALWAYS use the visual ruler method:

```
python -c "
header='[paste actual header line here]'
print('ENSDF 80-Column Ruler:')
print('Ones:
1234567890123456789012345678901234567890123456789012345678901234567890')
print('Tens:
11111111112222222222333333333334444444444555555555566666666667777777777888888888999
')
print('Header:', header)
print('Length:', len(header))
"
```

Process: Display 80-char ruler → Extract L/G records → Validate against ENSDF Manual → Report issues

"Debug Header Alignment"

IMMEDIATE ACTION: When header alignment issues are suspected:

1. Run `python .github/column_calibrate.py "filename" --header`
2. Compare with working reference files
3. Use the visual ruler technique to spot misalignments
4. Check ENSDF manual field positions (1-5, 6-9, 10-39, 40-65, 66-74, 75-80)

"What changed?"

MANDATORY FIRST STEP: Always run `git status` to identify ALL modified files.

Execute comprehensive change detection and documentation:

1. **FIRST:** Run `git status` to list all modified files
2. **Verify completeness:** Run `git diff --name-only HEAD` for cross-verification
3. **Check untracked files:** Run `git ls-files --others --exclude-standard`
4. **For each modified file:** Run `git diff HEAD~1 "filename"` to see what changed
5. **For moved files:** Use `git show HEAD~1:old/path/file` to examine previous content
6. **Update change.log** with evidence-based entries (never assume changes)
7. **Document with:**
 - Line numbers where changes occurred
 - Before/after content for significant changes
 - Scientific/technical context and rationale
 - File movement/reorganization details

PowerShell Considerations: Use `Select-Object -First N` instead of `head` for output limiting.

Remember: Git status MUST be the first step - missing files means incomplete documentation! Always cross-verify with multiple git commands to ensure complete coverage.

"Fix format!"

Auto-convert text to proper ENSDF notation:

Superscripts and Subscripts

- `{+n}` → superscript (e.g., `{+3}He` displays as ${}^3\text{He}$)
- `{-n}` → subscript (e.g., `H{-2}O` displays as H_2O)
- `{+-n}` → negative superscript (e.g., `10{+-4}` displays as 10^{-4})

Greek Letters and Mathematical Symbols

Greek lowercase:

- `|a` → α (alpha), `|b` → β (beta), `|c` → η (eta), `|d` → δ (delta)
- `|e` → ε (varepsilon), `|f` → φ (phi), `|g` → γ (gamma), `|h` → χ (chi)
- `|i` → ι (iota), `|j` → ϵ (epsilon), `|k` → κ (kappa), `|l` → λ (lambda)

- $|m \rightarrow \mu$ (mu), $|n \rightarrow \nu$ (nu), $|p \rightarrow \pi$ (pi), $|q \rightarrow \theta$ (theta)
- $|r \rightarrow \rho$ (rho), $|s \rightarrow \sigma$ (sigma), $|t \rightarrow \tau$ (tau), $|u \rightarrow \upsilon$ (upsilon)
- $|w \rightarrow \omega$ (omega), $|x \rightarrow \xi$ (xi), $|y \rightarrow \psi$ (psi), $|z \rightarrow \zeta$ (zeta)

Greek uppercase:

- $|D \rightarrow \Delta$ (Delta), $|F \rightarrow \Phi$ (Phi), $|G \rightarrow \Gamma$ (Gamma), $|L \rightarrow \Lambda$ (Lambda)
- $|P \rightarrow \Pi$ (Pi), $|Q \rightarrow \Theta$ (Theta), $|S \rightarrow \Sigma$ (Sigma), $|U \rightarrow \Upsilon$ (Upsilon)
- $|W \rightarrow \Omega$ (Omega), $|X \rightarrow \Xi$ (Xi), $|Y \rightarrow \Psi$ (Psi)

Mathematical symbols:

- $|* \rightarrow \times$ (times), $|? \rightarrow \approx$ (approx), $|< \rightarrow \leq$ (leq), $|> \rightarrow \geq$ (geq)
- $|' \rightarrow ^\circ$ (degree), $|+ \rightarrow \pm$ (plus-minus), $|- \rightarrow \mp$ (minus-plus)
- $|= \rightarrow \neq$ (not equal), $|@ \rightarrow \infty$ (infinity), $|^ \rightarrow \uparrow$ (up arrow)
- $|_ \rightarrow \downarrow$ (down arrow), $|& \rightarrow \equiv$ (equiv), $| (\rightarrow \leftarrow$ (left arrow)
- $|) \rightarrow \rightarrow$ (right arrow), $|\cdot \rightarrow \propto$ (proportional), $|| \rightarrow |$ (vertical bar)

Important Rules:

- For approximate values, use $|?$ (which gives both $\approx$ and $\sim$ symbols)
- Standalone $\sim$ is NOT allowed for approximate values in ENSDF

NEVER modify XREF lists during format fixes! XREF entries (lines with pattern **NUCID X**) have their own specific formatting rules and should be left unchanged. Only apply format fixes to comment text and other non-XREF content.

"Convert ENSDF to PDF"

Natural language request processing for ENSDF-to-PDF conversion using the enhanced **ens2pdf.py** script:

Example requests:

- "Convert S35_24mg_14n_3pg.ens to PDF"
- "Generate PDF from the adopted file"
- "Make PDF for the current ENSDF file"
- "ens2pdf for the current ens"
- "Convert Si35 files to PDF and open them"

Process: Automatically locates the specified .ens file, runs the Java conversion tool, and opens the resulting PDF

Script Usage:

```
# Convert single file by name
python ens2pdf.py Si35_adopted

# Convert with full file path
python ens2pdf.py "finished/Si35/new/Si35_adopted.ens"

# Convert all files for an element
```

```
python ens2pdf.py Si

# Convert files matching pattern
python ens2pdf.py "Si35_*sig"

# Convert and open in VS Code (default)
python ens2pdf.py Si35_adopted --open

# Convert and open in system viewer
python ens2pdf.py Si35_adopted --open --system
```

Features:

- **Smart PDF Opening:** Tries VS Code first, falls back to system viewer gracefully
- **Full Path Support:** Handles both relative names and complete file paths
- **Pattern Matching:** Use wildcards to convert multiple files
- **Cross-Platform:** Works on Windows, macOS, and Linux
- **Error Handling:** Graceful fallback when VS Code CLI tools aren't available
- **User Feedback:** Clear messages about conversion status and where PDF opened

PDF Location: All PDFs are generated in `D:/X/ND/Files/` directory

ENSDF Column Format Standards (CRITICAL - NO MISTAKES ALLOWED)

L-Record Format (Energy Levels):

Columns:
12345678901234567890123456789012345678901234567890123456789012345678901234567890
Format: 35XX L EEEE.E DE JP T DT L S
DS C
Example: 35P L 1572.0 1 1/2+ 2.29 PS 14 2 1.23
45

Field	Columns	Required	Description
NUCID	1-5	✓	Nucleus (e.g., "35P ")
CONT	6		Continuation flag
BLANK	7	✓	Must be blank
TYPE	8	✓	"L"
BLANK	9	✓	Must be blank
E	10-19	✓	Level energy (LEFT-JUSTIFIED)
DE	20-21		Energy uncertainty (LEFT-JUSTIFIED)
SPACE	22	✓	Readability space

Field	Columns	Required	Description
J	23-39		Spin-parity (LEFT-JUSTIFIED at col 23)
T	40-49		Half-life with units (LEFT-JUSTIFIED)
DT	50-55		Half-life uncertainty (LEFT-JUSTIFIED)
L	56-64		Angular momentum transfer
S	65-74		Spectroscopic strength
DS	75-76		Uncertainty in S (LEFT-JUSTIFIED)
C	77		Comment flag

G-Record Format (Gamma Transitions):

Columns:
12345678901234567890123456789012345678901234567890123456789012345678901234567890
Format: 35XX G EEEE.E DE II.I DI [M] MR DMR CC DCC TI
DTI C
Example: 35P G 1572.0 1 100.0 4 [E2] 1.23 0.45 0.0368 8 1.23
45

Field	Columns	Required	Description
NUCID	1-5	✓	Nucleus (e.g., "35P ")
CONT	6		Continuation flag
BLANK	7	✓	Must be blank
TYPE	8	✓	"G"
BLANK	9	✓	Must be blank
E	10-19	✓	Gamma energy (LEFT-JUSTIFIED)
DE	20-21		Energy uncertainty (LEFT-JUSTIFIED)
SPACE	22	✓	Readability space
RI	23-29		Relative photon intensity (LEFT-JUSTIFIED at col 23)
DRI	30-31		Uncertainty in RI (LEFT-JUSTIFIED, including GT, LT markers)
M	32-41		Multipolarity
MR	42-49		Mixing ratio
DMR	50-55		Uncertainty in MR (LEFT-JUSTIFIED)
CC	56-62		Conversion coefficient
DCC	63-64		Uncertainty in CC (LEFT-JUSTIFIED)

Field	Columns	Required	Description
TI	65-74		Total transition intensity
DTI	75-76		Uncertainty in TI (LEFT-JUSTIFIED)
C	77		Comment flag

Critical: ENSDF files are parsed by automated systems requiring exact positions. One column off = data rejection.

UNCERTAINTY LEFT-JUSTIFICATION RULE: ALL uncertainties (DE, DRI, DMR, DCC, DTI, DT, DS, etc.) MUST be left-justified in their respective fields, just like the values themselves. Special markers (GT, LT) within uncertainty fields are also left-justified.

LEFT-JUSTIFICATION RULE: ALL values AND uncertainties MUST be left-justified within their respective fields. This includes:

- Energy values (E field) and their uncertainties (DE field)
- J- π values (spin-parity) and any associated uncertainties
- Half-life values (T field) and their uncertainties (DT field)
- RI values (relative intensity) and their uncertainties (DRI field)
- Mixing ratios (MR field) and their uncertainties (DMR field)
- Conversion coefficients (CC field) and their uncertainties (DCC field)
- All numerical and text values AND their uncertainties

Never right-justify or center ANY values OR uncertainties in ENSDF records!

Essential Rules

File Protection

- **NEVER** edit `.old` files (reference files from previous evaluation rounds)
- **NEVER** modify first/last line indentation or spacing in `.ens` files
- **ALWAYS** preserve "PN" line with its numeric value
- Make all edits between first and last line boundaries only

ENSDF File Editing Safety Protocol

BEFORE ANY EDIT - MANDATORY CHECKS:

1. **Read current file state** - Never assume file structure
2. **Identify target line uniquely** - Must have 5+ lines of unique context
3. **Single field modification only** - Never edit multiple fields at once
4. **Validate column positions** - Check field boundaries before editing

EDITING METHODOLOGY:

1. **ONE EDIT AT A TIME** - Never batch multiple field changes
2. **PRECISE CONTEXT MATCHING** - Use complete L-record + surrounding context
3. **FIELD-SPECIFIC REPLACEMENTS** - Target only the specific field being changed

4. IMMEDIATE VALIDATION - Check file structure after each edit

EXAMPLE SAFE EDIT PATTERN:

Target: Change T field value from "0.025 EV 1" to "0.027 EV 2" in line with energy 34.03

CORRECT approach:

- Read file to confirm current state
- Use complete L-record as context: " 35S L 34.03 1 1/2 0.025 EV 1 (2) 34.03 1 "
- Replace only T field portion: "0.025 EV 1" → "0.027 EV 2"
- Validate file structure immediately

WRONG approach:

- Assume file state
- Edit multiple fields simultaneously
- Use insufficient context
- Continue editing after any error

CRITICAL: If any edit causes file corruption, STOP immediately and inform user

Column Positioning

- **J- π placement:** Always start at column 23, LEFT-JUSTIFIED (never add spaces that shift uncertainties)
- **Energy values:** LEFT-JUSTIFIED in their designated columns (10-19)
- **RI values:** Start at column 23, **LEFT-JUSTIFIED** in 7-char field (23-29)
- **DRI values:** Position at columns 30-31 (including special markers like GT, LT)
- **Half-life values:** LEFT-JUSTIFIED in T field (columns 40-49)
- **BR values:** Position at column 32 (N-records), LEFT-JUSTIFIED
- **NR values:** Columns 11-15 (N-records), LEFT-JUSTIFIED

CRITICAL: ALL values must be LEFT-JUSTIFIED within their respective fields - never right-justified or centered!

⚠ CRITICAL COLUMN RULE: When fixing a quantity's position to the correct columns, NEVER shift other field values to wrong columns!

- L-transfer values: Must stay in columns 56-64
- Spectroscopic factors: Must stay in columns 65-74
- Comment flags: Must stay in column 77
- Only adjust spacing between fields - never move field data to incorrect columns!

NSR Keynumber Formatting

- **In comments/records:** Second letter lowercase (2023Bo17, 2021Wa16)
- **In headers/Q-records:** All uppercase (2023B017, 2021WA16)

Change Tracking

- **Always** update [.github/change.log](#) after significant changes

- **Never** create duplicate change.log files
- Use evidence-based documentation with specific line numbers
- **Never** document assumed changes - always verify with tools

ENSDF Special Characters

Superscripts/Subscripts

- $\{+n\} \rightarrow$ superscript (e.g., $\{+35\}\text{Ar} \rightarrow {}^{35}\text{Ar}$)
- $\{-n\} \rightarrow$ subscript (e.g., $\text{T}\{-1/2\} \rightarrow \text{T}_{1/2}$)
- $\{+-n\} \rightarrow$ negative superscript (e.g., $\{+-4\} \rightarrow {}^{-4}$)

Greek Letters

Lowercase: $|a| \rightarrow \alpha$, $|b| \rightarrow \beta$, $|g| \rightarrow \gamma$, $|d| \rightarrow \delta$, $|e| \rightarrow \epsilon$, $|l| \rightarrow \lambda$, $|m| \rightarrow \mu$, $|n| \rightarrow \nu$, $|p| \rightarrow \pi$, $|r| \rightarrow \rho$, $|s| \rightarrow \sigma$, $|t| \rightarrow \tau$, $|w| \rightarrow \omega$ **Uppercase:** $|D| \rightarrow \Delta$, $|G| \rightarrow \Gamma$, $|L| \rightarrow \Lambda$, $|P| \rightarrow \Pi$, $|S| \rightarrow \Sigma$, $|W| \rightarrow \Omega$

Mathematical Symbols

- $|*| \rightarrow \times$ (times), $|?| \rightarrow \approx$ (approx), $|+| \rightarrow \pm$ (plus-minus), $|-| \rightarrow \mp$ (minus-plus)
- $|<| \rightarrow \leq$, $|>| \rightarrow \geq$, $|'| \rightarrow ^\circ$, $|=| \rightarrow \neq$, $|@| \rightarrow \infty$
- $|\wedge| \rightarrow \uparrow$, $|_|| \rightarrow \downarrow$, $|(|)| \rightarrow \leftarrow$, $|)| \rightarrow \rightarrow$, $|.| \rightarrow \propto$, $||| \rightarrow |$

Important: Use $|?|$ for approximate values, never standalone $\sim$ (except in names/mass notation)

Common Examples

- $\%(|e+|b\{++\})p \rightarrow \%(\epsilon+\beta^+)p$
- $\{+208\}\text{Pb}(\{+36\}\text{S}, \{+35\}\text{S}) \rightarrow {}^{208}\text{Pb}({}^{36}\text{S}, {}^{35}\text{S})$
- $|s(E(\{+3\}\text{He}), |q)| \rightarrow \sigma(E({}^3\text{He}), \theta)$

Academic Standards

Citation Tense

Use PAST tense for all references to completed studies:

- ✓ "Authors stated...", "1994FO04 measured...", "Previous evaluators concluded..."
- X "Authors state...", "1994FO04 measures..."

Grammar Fixes

Common corrections: "stoped" $\rightarrow$ "stopped", "usign" $\rightarrow$ "using", "coefficients" $\rightarrow$ "coefficients"

Nuclear Data Evaluation

General Comment Ordering (adopted.ens files)

1. **Isotope discovery** (reference): experimental details
2. **$\{+A\}\text{X}$ production**: production methods and studies
3. **$\{+A\}\text{X}$ decay measurements**: half-life, decay modes

4. **{+A}X radius measurement:** nuclear radius determinations
5. **{+A}X mass measurements:** mass spectrometry, Q-values
6. **Theoretical calculations:** models, predictions (always last)

L-Transfer Rules for J- π Assignment

- $L=0 \rightarrow J-\pi$: 1/2+
- $L=1 \rightarrow J-\pi$: 1/2-, 3/2-
- $L=2 \rightarrow J-\pi$: 3/2+, 5/2+
- $L=3 \rightarrow J-\pi$: 5/2-, 7/2-

Note: Always confirm with experimental data; never enter L-values in J- π column.

Tools and Workflows

PDF Generation

```
# Single element
Set-Location "D:\X\ND\Files"
$element = "Al"
Get-ChildItem "D:\X\ND\A35\finished\${element}35\new\*.ens" | ForEach-Object {
    java -jar "D:\X\ND\McMaster-MSU-Java-
NDS\McMaster_MSU_JAVA_NDS_v3.0_01May2025.jar" $_.FullName "$($_.BaseName).pdf"
}

# All elements
$elements = @("Al", "Ar", "Ca", "K", "Mg", "Na", "Ne", "P", "Si")
foreach ($element in $elements) {
    Get-ChildItem "D:\X\ND\A35\finished\${element}35\new\*adopted.ens" | ForEach-
Object {
        java -jar "D:\X\ND\McMaster-MSU-Java-
NDS\McMaster_MSU_JAVA_NDS_v3.0_01May2025.jar" $_.FullName "$($_.BaseName).pdf"
    }
}
```

Change Detection Process

1. **Pre-work (MANDATORY):** `git status`, `git diff --name-only HEAD`
2. **During work:** Track file modifications systematically
3. **Post-work:** Use all detection tools on ALL files from git status
4. **Documentation:** Evidence-based change.log entries with line numbers

CRITICAL REMINDER: Always start with `git status` - this shows the complete picture!

File Categories to Track

- **ENSDF source files:** *.ens files (most important)
- **Generated PDFs:** *.pdf files (expected to change when source changes)
- **Processing artifacts:** temp/. files (expected, document but don't commit)

- **Tools and scripts:** .github/. files (important for tooling changes)
- **Documentation:** README.md, change.log, etc.

Evidence-Based Documentation Rules

Every change log entry should be backed by:

- Specific file diffs from `git diff HEAD~1 "filename"`
- Line numbers where changes occurred
- Actual before/after content when significant
- Explanation of why the change was made

Key principle: Use multiple detection methods and always cross-verify. If git shows a file changed, dig deeper with git diff. If you modified an ENSDF file, expect to see corresponding PDF changes.

Verification Checklist

- ☐ **FIRST:** `git status` - identify ALL modified files (MANDATORY)
- ☐ `git diff --name-only HEAD` - complete list verification
- ☐ `git ls-files --others --exclude-standard` - untracked files
- ☐ `git diff HEAD~1 "filename"` on each modified file from git status
- ☐ For moved files: `git show HEAD~1:old/path/file | Select-Object -First 20` (PowerShell)
- ☐ For large outputs: Use `Select-Object -First N` to limit output in PowerShell
- ☐ Update `change.log` with evidence-based entries
- ☐ Document file movements/reorganizations with full context
- ☐ Comprehensive commit message
- ☐ Cross-check: did any ENSDF changes result in expected PDF updates?

Remember: Start every workflow with git status and use PowerShell-compatible commands!

Git Commit Template

Title: Brief description of main changes

Summary:

- Enhanced/improved/fixed major components
- Scientific content updates in specific files

ENSDF Tools:

- tool_name.py: Specific improvements and validation results

Scientific Content:

- file_name.ens: Changes with line numbers and rationale

Processing Artifacts:

- PDF files: Regenerated files listed
- Temp files: Expected analysis output updates

Files changed: X modified, Y untracked

Brief scope and impact summary

Example Commit Structure

Title: Enhance ENSDF column calibration tools and improve Ar35 scientific content

Summary:

- Enhanced Python column calibration script with complete 80-column ENSDF format support
- Improved scientific content and formatting in Ar35 ENSDF files
- Completed comprehensive change tracking and documentation

ENSDF Tools:

- `column_calibrate.py`: Extended from 41-column to complete 80-column ENSDF support
- `check_averages.py`: Completed and tested average calculation verification tool

Scientific Content:

- `Ar35_36ar_p_d.ens`: Fixed grammar in L=3 vs L=2 comparison (line 77)
- `Ar35_adopted.ens`: Multiple scientific and formatting enhancements

Processing Artifacts:

- PDF files: Regenerated `Ar35_36ar_3he_a.pdf`, `Ar35_36ar_p_d.pdf`, `Ar35_adopted.pdf`
- Temp files: Updated all analysis outputs (`35.err`, `35.fed`, `35.fmt`, etc.)

Files changed: 15 modified, 2 untracked

Completion of comprehensive ENSDF column calibration tooling and systematic improvement of Ar35 nuclear data content.

Project Structure

Core Files (Most Critical)

- `finished/[Element]/new/*.ens` - Primary ENSDF source files
- `.github/change.log` - Comprehensive change tracking

Generated Files (Expected to Change)

- `finished/[Element]/pdf/*.pdf` - Generated from `.ens` files
- `finished/[Element]/temp/*.*` - Analysis tool artifacts

Tools

- `ens2pdf.py` - Python script for automated ENSDF to PDF conversion
- `.github/column-calibrate.ps1` - PowerShell column validator
- `.github/column_calibrate.py` - Python column validator
- `.github/check_averages.py` - Average calculation validator

Reference Files (NEVER EDIT)

- `*.old` files - Previous evaluation rounds, keep untouched

Focus Areas

Current Priority: K35 and P35 files (Ar35 completed)

Quality Assurance: Use Self-Calibrate Columns before any ENSDF edits, use What changed? after any modifications

Remember: Nuclear data accuracy is critical - when in doubt, verify with tools and cross-check against ENSDF Manual specifications.

Image Data Extraction Protocol

Level Scheme Analysis

- **Systematic scanning:** Left-to-right, top-to-bottom approach
- **Energy identification:** Clear notation for parentheses, uncertainties, tentative assignments
- **Color coding:** Black (known) vs Red (new) vs other markings
- **Special notations:** Asterisks (*), question marks (?), parentheses ()
- **Cross-verification:** Compare extracted data with tabulated lists

Spectral Analysis

- **Peak identification:** Exact energy labels, not estimates
- **Gate verification:** Check coincidence logic with nuclear structure
- **Contamination markers:** Identify non-target nuclide peaks
- **Quality indicators:** Intensity, resolution, background

Quality Control

- **Never guess or interpolate** energy values
- **Admit uncertainty** when image quality is poor
- **Section-by-section verification** before final compilation
- **Cross-check** with provided data tables

DCO Ratio and Polarization Analysis

Essential for multipolarity assignments in gamma-ray spectroscopy

DCO Ratio Rules

- **DCO(D) ≈ 1.0** → Dipole transition (M1, E1, or M1+E2 with dominant M1)
- **DCO(D) ≈ 1.6** → Quadrupole transition (E2 or M2)
- **DCO(Q) ≈ 1.0** → Quadrupole transition (E2 or M2)
- **DCO(Q) ≈ 0.6** → Dipole transition (M1, E1, or M1+E2 with dominant M1)

Polarization Rules

- **POL > 0** → Electric transition (E1, E2, etc.)
- **POL < 0** → Magnetic transition (M1, M2, etc.)
- **POL ≈ 0** → Mixed transition or measurement uncertainty

Quality Control Guidelines

- **Expected DCO ranges:** 0.4-1.4 for dipole, 0.8-1.8 for quadrupole
- **Red flags:** DCO > 2.0 or DCO < 0.3 (possible contamination or experimental issues)
- **Borderline values:** 0.8-1.2 may require additional analysis
- **Cross-verification:** Always check DCO consistency with nuclear structure logic

Systematic Analysis Protocol

1. **Extract all DCO and POL data** from experimental comments
2. **Apply rules systematically** to each transition
3. **Identify inconsistencies** between assigned multipolarity and DCO/POL
4. **Flag unusual values** (DCO > 2.0) for further investigation
5. **Document findings** with specific energy, DCO value, and recommended assignment

Image Data Extraction Request

You are an expert nuclear data scientist with extensive experience handling ENSDF-formatted data. Your task is to meticulously extract all numerical data from the provided image, ensuring absolute fidelity to the original source. Preserve every decimal place exactly. Do not round, omit, alter, or add any digits. For example, 10.0 is 10.0, not 10 or 10.00!

To ensure proper column alignment, please utilize a null value for any empty fields. It is important to avoid misinterpreting other fields or fabricating placeholder values to fill these unfilled spaces.

Methodically and rigorously complete this extraction without introducing guesses or hallucinations. Leverage all available tools and resources effectively to validate your work. Double-check all values at least once before finalizing your response. Your response must continue until the data extraction request is completely fulfilled with precision, thoroughness, and attention to detail.

Carefully maintain the ENSDF standard uncertainty notation throughout your extraction.

The uncertainty digits align precisely with the rightmost decimal digit of the stated value per ENSDF standards: ENSDF Uncertainty Notation (Clear Examples) Decimal Digits ENSDF Notation Meaning (explicit $\pm$ form)

No decimal: 1234(5) 1234 $\pm$ 5 1234(56) 1234 $\pm$ 56 1234(567) 1234 $\pm$ 567 1 decimal: 12.3(4) 12.3 $\pm$ 0.4 12.3(45) 12.3 $\pm$ 0.45 12.3(456) 12.3 $\pm$ 0.456 2 decimals: 1.23(4) 1.23 $\pm$ 0.04 1.23(45) 1.23 $\pm$ 0.045 1.23(456) 1.23 $\pm$ 0.0456 3 decimals: 0.123(4) 0.123 $\pm$ 0.004 0.123(45) 0.123 $\pm$ 0.0045 0.123(456) 0.123 $\pm$ 0.00456 4 decimals: 0.0123(4) 0.0123 $\pm$ 0.0004 0.0123(45) 0.0123 $\pm$ 0.00045 0.0123(456) 0.0123 $\pm$ 0.000456